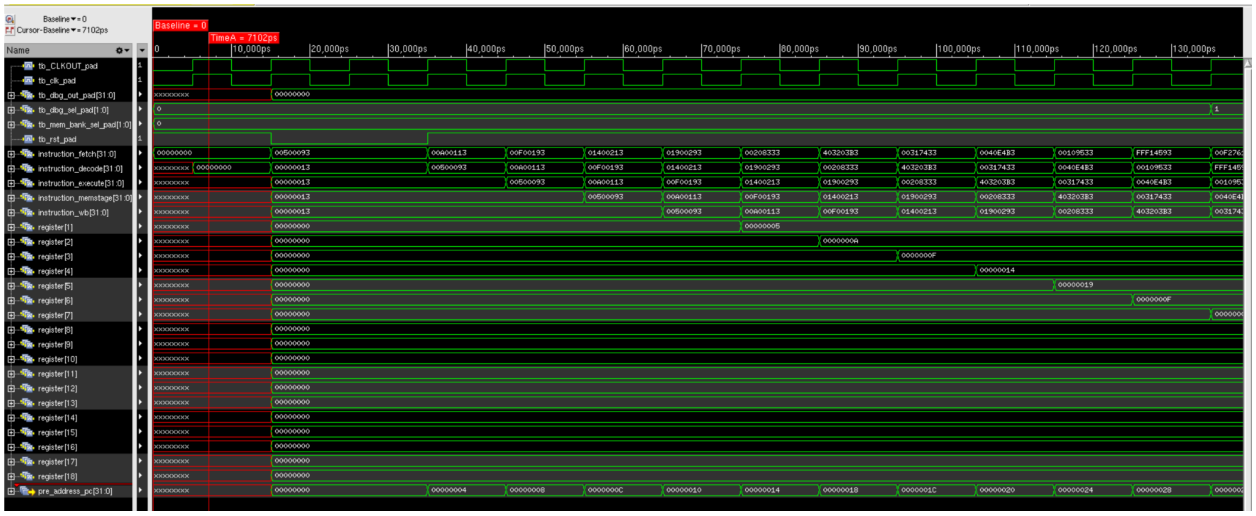
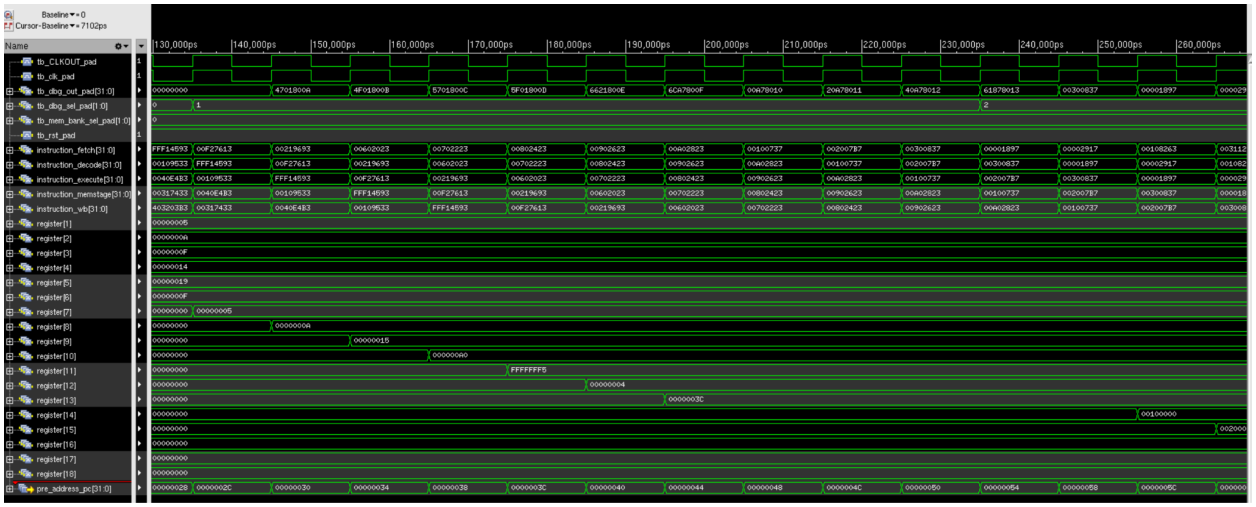
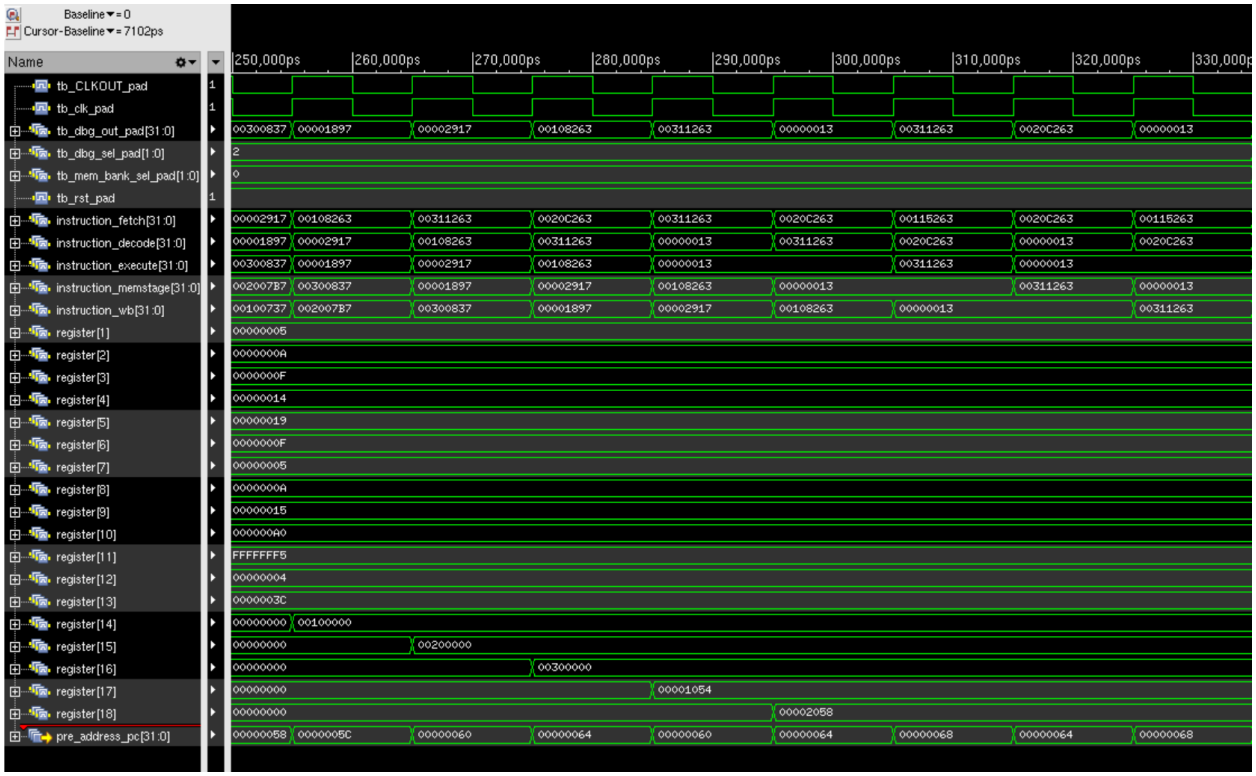
Simulation of 5 instructions of each type

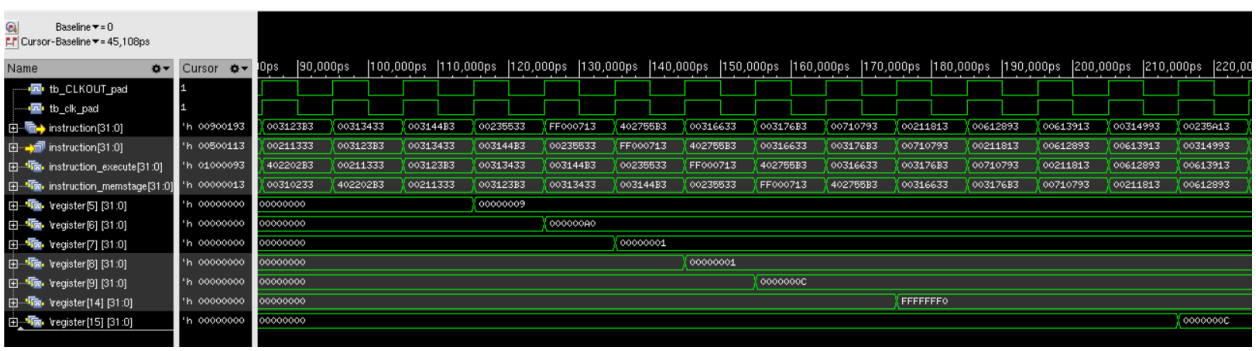
```
// 1. INITIALIZATION
00500093 // addi x1, x0, 5   (x1 = 5)
00A00113 // addi x2, x0, 10  (x2 = 10)
00F00193 // addi x3, x0, 15  (x3 = 15)
01400213 // addi x4, x0, 20  (x4 = 20)
01900293 // addi x5, x0, 25  (x5 = 25)

// 2. R-Type Instructions
00208333 // add x6, x1, x2   (x6 = 5 + 10 = 15)
403203B3 // sub x7, x4, x3   (x7 = 20 - 15 = 5)
00317433 // and x8, x2, x3   (x8 = 10 & 15 = 10)
0040E4B3 // or  x9, x1, x4   (x9 = 5 | 20 = 21)
00109533 // sll x10, x1, x1  (x10 = 5 << 5 = 160)

// 3. I-Type Instructions (Immediate Logic)
FFF14593 // xori x11, x2, -1 (x11 = 10 ^ -1 = -11)
00F27613 // andi x12, x4, 15 (x12 = 20 & 15 = 4)
00219693 // slli x13, x3, 2  (x13 = 15 << 2 = 60)

// 4. S-Type Instructions (Store to Memory)
// We will store our calculated values to base address 0 (using x0)
00602023 // sw x6, 0(x0)     (Store Word: x6 [15] at addr 0)
00702223 // sw x7, 4(x0)     (Store Word: x7 [5]  at addr 4)
00802423 // sw x8, 8(x0)     (Store Word: x8 [10] at addr 8)
00902623 // sw x9, 12(x0)    (Store Word: x9 [21] at addr 12)
00A02823 // sw x10, 16(x0)   (Store Word: x10 [160] at addr 16)

// 5. U-Type Instructions (Upper Immediate)
00100737 // lui   x14, 0x100 (x14 = 0x100 << 12 = 0x100000)
002007B7 // lui   x15, 0x200 (x15 = 0x200 << 12 = 0x200000)
00300837 // lui   x16, 0x300 (x16 = 0x300 << 12 = 0x300000)
00001897 // auipc x17, 0x1   (x17 = PC + (0x1 << 12))
00002917 // auipc x18, 0x2   (x18 = PC + (0x2 << 12))

// 6. B-Type Instructions (Branch / Control Flow)
// These branch to PC+4 (the next instruction) so execution continues safely
00108263 // beq  x1, x1, +4  (Branch if x1 == x1 [5 == 5] to PC+4)
00311263 // bne  x2, x3, +4  (Branch if x2 != x3 [10 != 15] to PC+4)
0020C263 // blt  x1, x2, +4  (Branch if x1 < x2  [5 < 10] to PC+4)
00115263 // bge  x2, x1, +4  (Branch if x2 >= x1 [10 >= 5] to PC+4)
00106263 // bltu x0, x1, +4  (Branch if x0 < x1 unsigned [0 < 5] to PC+4)
```

Screenshots from the waveform viewer









We have removed the design files from the GLS_Simulation folder. While running, we need to add those.